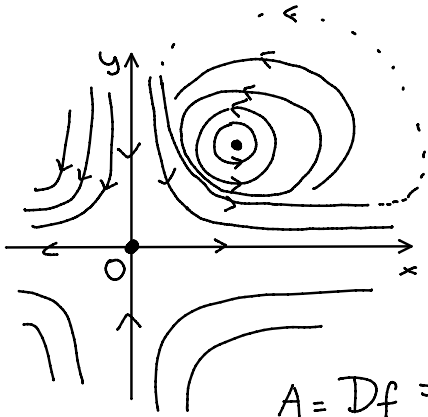


Lotka-Volterra model



$$\dot{x} = \alpha x - \beta xy = f_1$$

$$\dot{y} = -\gamma y + \delta xy = f_2$$

fixed points

$$\begin{cases} x(\alpha - \beta y) = 0 \rightarrow x=0 \text{ or } y = \alpha/\beta \\ y(-\gamma + \delta x) = 0 \rightarrow y=0 \text{ or } x = \gamma/\delta \end{cases}$$

$$A = Df = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} \alpha - \beta y & -\beta x \\ \delta y & -\gamma + \delta x \end{bmatrix}$$

Th: 1) $[Df]_O = \begin{bmatrix} \alpha & 0 \\ 0 & -\gamma \end{bmatrix} \leadsto \dot{x} = [Df]_O x \rightarrow \begin{cases} \dot{x} = \alpha x \\ \dot{y} = -\gamma y \end{cases} \rightarrow \begin{cases} x(t) = c_1 e^{\alpha t} \\ y(t) = c_2 e^{-\gamma t} \end{cases}$

2) $[Df]_B = \begin{bmatrix} 0 & -\frac{\delta \alpha}{\beta} \\ \frac{\delta \alpha}{\beta} & 0 \end{bmatrix} \rightarrow \begin{cases} \dot{x} = -d_1 y \\ \dot{y} = d_2 x \end{cases}$ (try this as a homework)

$$\dot{x}_1 = \alpha x_1 - \gamma x_1 x_2$$

$$\frac{1}{x_1} \dot{x}_1 = \alpha - \gamma x_2 - \delta x_3 - \gamma x_4$$